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Stepwise Noise Elimination for Better Motivational and Advisory Texts Classification

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There is little research into designing artificial motivational agents. The end-goal of our studies is therefore to create a dialogue system that would motivate users to do their everyday tasks using natural language. In this paper, we present a method of distinguishing texts containing motivational advice from regular texts to sort out noise in training data for our dialogue system. We implemented a novel method of chaining two shallow networks together by utilizing the output results of the first network to determine the input for the second one. We achieved F-score of 0.94 and 0.97 with our proposed method. The contributions of this paper are threefold: first, we successfully identified 14 hand-crafted features that make a text motivational/advisory. Secondly, we were able to create a classifying algorithm that distinguishes motivational/advisory texts from regular ones. Finally, our proposed method can be applied to other text classification tasks.

Keywords: text classification, motivation, advisory systems, neural networks, dialogue systems

1. Introduction

The concept of motivation has been extensively studied in psychology, but not so much in the field of dialogue systems. However, a personal motivational agent can be of much help to people who have problems with managing their activities or who, for various reasons, feel unmotivated. In this paper, we describe the first stage of research aiming towards creating a dialogue system able to motivate its users to perform their everyday tasks.

Motivation is an important contemporary issue, mostly researched within the context of professional life [1–3], academic life [4, 5] or mental health issues, usually mood disorders [6, 7]. There are numerous websites specifically targeted at people who feel the lack of motivation in their lives and search online for help (for example [8, 9]). The online discussion platform Reddit: Get Disciplined [10], which defines its purpose as *Help[ing] others attain self-discipline by sharing what helps [the user]*, has as much as over 360,000 subscribers as of November 2018. All this shows the magnitude of the problem of being unmotivated and the need and desire to solve that problem. Our dialogue system can be of help to many people struggling with such issues in their everyday lives.

The end goal of our research is to create a dialogue system that would motivate the user to complete tasks on their schedule, regardless of the type of task or reason for being unmotivated, by giving motivational advice expressed in natural language. The input to the system is the user's description about their lack of motivation, and the output is a response meant to give the user some advice pertaining to the problem at hand. In this research, we define motivational advice as *an utterance in natural language pertaining to the user's problem with being unmotivated and containing a possible solution to that problem.* The target users are therefore people struggling with motivation in their everyday life.

Such a dialogue system will need an enormous amount of training data to be able to produce its own utterances. This data is not readily available at any given moment, since there is no corpus of such texts. The data can be retrieved from various different sources instead, for example online discussion platforms or motivational blogs, but in that case, it is crucial to eliminate noise, which may constitute a large portion of the data. Therefore, in this paper we describe an algorithm that recognizes motivational texts and distinguishes them from non-motivational ones. The algorithm is meant to sort out texts that do not contain motivational advice. The texts that the algorithm labels as motivational can then be analyzed further and fed as training data into our final dialogue system.

The most important contribution of this paper is our original method of text classification that we developed for our task. Another novel point of this research is that currently there is no algorithm distinguishing advisory and/or motivational utterances from regular ones. Moreover, to the best of our knowledge, ours is the only project currently in development which will result in a dialogue system that motivates people to perform all kinds of tasks using natural language, imitating an actual human-to-human conversation.

The structure of this paper is as follows. First, we characterize the data that we used (Section 2.1) and our 14 advice features (Section 2.2). Next, we describe our preliminary experiments using both shallow and deep neural

networks (Section 3). Section 4 contains a characterization of our proposed method, with network architecture (Section 4.1), experiment details and results (Sections 4.2 and 4.3), comparison with a related method (Section 4.4) and a discussion about the proposed method, its possible future modifications and applications (Section 4.5). Section 5 presents related work in both user motivation and neural networks for classification problems. Section 6 concludes the paper with a review of the findings from the paper and their significance for detecting motivating/advisory texts.

2. Datasets and Features

2.1. Datasets

In our study, we analyzed data from an online discussion platform Reddit. It is a website divided thematically into various so-called subreddits, where users can exchange opinions and information about given topics, and the data can be downloaded freely for research purposes. The subreddit *r/getdisciplined* [10] is a place for unmotivated people to seek motivational advice from others. Concretely, a user submits a post describing his or her problem with motivation, and other users offer suggestions on how to deal with the issue. The posts cover a broad range of topics, from being unmotivated at work or university to procrastination stemming from mental health issues. As indicated in community rules, people who comment are required to *help others attain self-discipline, by sharing what helps [them].* We chose this subreddit because posts and comments represent exactly the type of input and output, respectively, that we would like to have in our dialogue system. To ensure that there will be no mislabeling problem – essentially, that all comments downloaded from this subreddit would actually be motivational – we downloaded only 3 top comments from posts that had more than five comments in total.

Both posts and comments can be voted on by other Reddit users. Therefore, the quality of motivational advice expressed in a comment can be measured with the amount of upvotes and downvotes that the comment received. In other words, the best rated comments were considered the most motivating by the community. Ideally, we would like to measure this quality by seeing whether the original poster upvoted the comment, but Reddit does not provide such information. Relying on the voting system has the additional benefit of crowdsourcing the ratings, which allowed us to exploit intersubjectivity among numerous Reddit users instead of only a few annotators.

Our initial analysis of motivational comments using sentiment, polarity and cosine similarity between posts and comments revealed that simple surface features will not be helpful in determining the motivational level of a text, or the quality of advice expressed in that text. Therefore, we performed a short qualitative analysis on the best ranked comments from *r/getdisciplined* and established a list of semantic features separating them from other comments. These features will be described in Section 2.2 below.

Table 1. Data for the classification experiments.

Class	Subreddit	Number of comments
Motivational	*r/getdisciplined*	889
Non-motivational	*r/todayilearned*	890
	Total	1,779

As negative examples, we downloaded regular comments from the subreddits *r/todayilearned* [11], which is dedicated to discussions about interesting pictures and links, and therefore unlikely to contain any advice at all. Our goal was to train a classifier to differentiate between the motivational/advisory comments and regular ones. **Table 1** shows our data.

Unfortunately, we were not able to obtain more data due to the low activity level on the motivational subreddit. Using Reddit API [12] limits the amount of data that can be downloaded at any given time to around 1,000 latest posts, so we were unable to use older posts and comments. Even though the *r/todayilearned* subreddit is much more active, we opted to download the same amount of data from there to have a balanced dataset.

2.2. Features

Our quality analysis revealed that the best ranked comments from *r/getdisciplined* usually had two things in common: they provided very specific, practical advice for the troubled user who made the post, and included expressions of being able to relate to the struggles of that user, usually because the commenter had to deal with the same problem in the past. We operationalized these characteristics into several text features in our experiment. Overall, we decided on 14 distinct features, which will be described below.

Before analyzing the comments, we pre-processed each one by detecting sentence boundaries, assigning part-of-speech tags, and, for some feature calculations, removing stopwords. From now on, we will use names *wordlist_withstops* for a list of all words in the comment, *wordlist_nostops* for the same list with stopwords removed, and *sent_list* for a list of sentences in the comment. We also removed any non-ASCII characters. This means that we did not process any special characters such as emoticons.

The features were as follows:

Sentics Scores of **aptitude**, **attention**, **pleasantness**, and **sensitivity** measured using the Sentic library for Python on *wordlist_nostops*. The library is an API to the SenticNet knowledge base [13], which contains information on sentiment values of words. All the sentics values fall on the scale between -1 and 1.

Relatability Score measured by the percentage of first person pronouns (including possessive pronouns) in *wordlist_withstops*. The Score range was 0 to 1.

Imperative Score measured by the percentage of imperative expressions in the comment text. Specifically, we looked for clauses beginning with non-infinitive verbs, the word *please* preceding a verb, the phrase *why don't you* and phrases comprised of *you* or *OP* (*Original Poster*, which is a popular way of referring to the author of the post on Reddit) and a modal verb. We excluded sentences that started with auxiliary verbs *do* and *have* to avoid counting in questions with syntax similar to imperative expressions. Since most of these expressions are bigrams, we counted the percentage on number of all words divided by 2. The Score range was 0 to 1.

Advice Score defined as the count of advisory expressions in the text. For this purpose, we prepared a list of possible advisory expressions: *you need to*, *op needs to*, *you have to*, *op has to*, *it might be worth*, *I would*, *it would be good to*, *it might be good to*, *you had better*, *you'd better*, *your only option is*, *why don't you*, *have you thought about*, *have you tried*, *how about*, *if I were you*, *recommend*, *suggest*, *advise*, *you could always*, *have you considered*, *why not*. This feature also counted website links, which were a way of offering advice in many comments in our datasets. Therefore, in the preprocessing stage we replaced all links with the token *[link]*, which was then also used by the Advice Score as an advisory expression. The overall comment score was divided by 10 so that we could scale it down to range close to 0–1. Having features in this range is good practice in deep learning and facilitates the training process.

Specificity Scores including six features. Specificity score was first proposed by [14] to help extract suggestions and complaints from employee surveys and online product reviews. The goal was to find sentences containing specific content, which we adapted in our experiment for the purpose of finding specific motivational advice in Reddit comments.

We calculated the scores as described in [14] with slight modifications, which we will detail below. The calculations were performed for each sentence in the comment using *sent_list*, and the final scores for the entire comment were obtained by adding up all the sentence scores. The features were: **Average Semantic Depth** (ASD), **Average Semantic Height** (ASH), **Total Occurrence Count** (TOC), **Count of Named Entities** (CNE), **Count of Proper Nouns** (CPN), and **Sentence Length** (LEN).

Features like ASD, ASH, and TOC required access to the WordNet ontology [15]. For ASD and ASH, we had to retrieve hypernymy and hyponymy hierarchies for each content word, respectively. For ASD, we determined the longest path in the hypernymy hierarchy that led from the word to the top of the hierarchy (see **Fig. 1**). The length of this path determined the Semantic Depth of the word in this hierarchy. For ASH, we performed analogical calculations with the hyponymy hierarchy – we determined the shortest path from the word to its lowest hyponym. This path represents the Semantic Height of the word in the hierarchy. To obtain the ASD score for the entire sentence, we added all the ASDs for all the content words (meaning nouns, verbs, adjectives, and adverbs) and divided the sum by the total number of content words in the sentence (by which we mean a sentence from *sent_list* with stopwords removed). We performed the same calculations for ASHs. For TOC, the number of times a word occurs in the ontology is stored in the WordNet database and can be easily obtained. We measured it by obtaining occurrence count in WordNet for each lemmatized content word and adding up three lowest scores in a sentence.

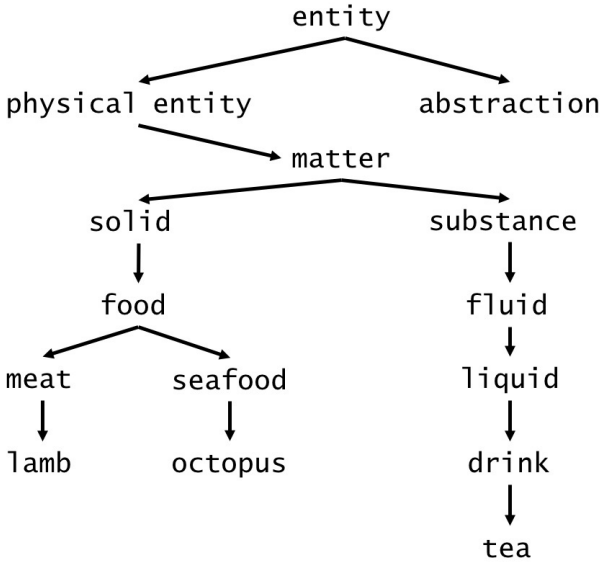


Fig. 1. The hypernymy/hyponymy hierarchy in WordNet based on [16]. In this example, the Semantic Depth of the word *seafood* is 5, because it takes five steps to reach the top of the hierarchy (the *entity* semantic element). Likewise, the Semantic Height of the word *substance* is 4, because it takes four steps to reach the lowest hyponym under that word.

Count of Named Entities (CNE) meant the number of named entities in the sentence that included stopwords, determined with a NLTK [17] Named Entities tagger. Count of Proper Nouns (CPN) was measured by the number of proper nouns in the sentence with stopwords removed. Specifically, we looked for part-of-speech tags NNP, NNPS, and CD. The first two mark singular and plural proper nouns, respectively. The CD tag marks cardinal numbers. This is technically not a proper noun, but was included in CPN calculations in [14], because according to the authors, sentences containing numbers tend to be more specific. We decided to keep our calculations the same. The last Specificity Score, Sentence Length, was the number of words in the sentence with stopwords removed.

There were several modifications that we introduced compared to [14]. The first one was changes in calculating ASH: while the authors of [14] counted the *longest* path to the lowest hyponym, we decided to count the *shortest* path, as we assumed it would be more indicative of how specific the word is. The second one was that we omitted converting every content word into a noun. The authors of

Table 2. Example scores for a motivational comment.

Comment						
Just take it one day at a time. What can you do right now to improve your life? Clean your room? Make your bed? Go outside? Work out? Read? Anything really. Start setting small goals for yourself and just take it one day at a time. What is it you want to accomplish? You don't need to know exactly but you need to know so you can start working toward it.						
Aptitude	Attention	Pleasantness	Sensitivity	Relatability Score	Imperative Score	Advice Score
0.08	0.02	0.16	0.13	0.00	0.12	0.1
ASD	ASH	ASHD	TOC	CNE	CPN	LEN
4.18	3.99	0.19	0.25	0.20	0.20	0.36

the paper performed this step because at the time of their research the hypernymy/hyponymy hierarchy in WordNet was relatively complex for nouns but very sparse for verbs and adjectives. We felt that currently there is no need for such modifications, as WordNet has grown significantly since then. Finally, we did not combine the features into one Specificity Score, but we used them all separately in our algorithm.

We also introduced one additional specificity feature. We combined Average Semantic Depth and Average Semantic Height into an additional ASHD feature by deducting the overall ASH score from the overall ASD score for each comment. This was done to reflect the difference between the two features, which proved to be important after careful error analysis of our experiments described in [18].

We then divided ASD, ASH, ASHD, TOC, and LEN by 100 and CNE and CPN by 10 to put the scores in the same numerical range as other features. Before division, ASD, ASH, TOC had rather high values that went into dozens. This is because the scores were calculated on a sentence-to-sentence basis and then added together to obtain the score of the entire text. For example, if there were three sentences in the text with ASD of 4, 5, and 3.5 respectively, the text score was $4 + 5 + 3.5 = 12.5$. We divided these scores by 100 to put them closer to the 0–1 scale. The same goes for LEN, which was the length of the entire text and also tended to have high values before division. ASHD was divided by 100 so that it would properly reflect the difference between the scaled-down ASD and ASH. On the other hand, CNE and CPN meant the count of named entities and proper nouns in the text, which tended to be much lower overall, so we only divided them by 10 to scale them down. It should be mentioned here that this scaling does not guarantee that the features will actually get into the 0–1 range; the goal was merely to approximate them to those numbers to help the neural network in training.

All in all, we had 14 features in our modified feature set: the four Sentics Scores, Relatability Score, Imperative Score, Advice Score, Average Semantic Depth, Average Semantic Height, ASHD being the difference between Average Semantic Depth and Average Semantic Height, Total Occurrence Count, Count of Named Entities, Count of Proper Nouns and Sentence Length.

Table 3. Feature scores for the *r/getdisciplined* versus *r/todayilearned* classification task. We bolded mean differences between both categories that stand out.

		r/getdisciplined	*r/todayilearned*
Aptitude	Mean	**0.139**	**0.098**
	Median	0.142	0.089
Attention	Mean	0.079	0.074
	Median	0.077	0.053
Pleasantness	Mean	**0.114**	**0.082**
	Median	0.115	0.045
Sensitivity	Mean	0.062	0.049
	Median	0.058	0.000
Relatability Score	Mean	0.039	0.033
	Median	0.027	0.000
Imperative Score	Mean	0.063	0.023
	Median	0.052	0.000
Advice Score	Mean	**0.045**	**0.002**
	Median	0.000	0.000
ASD	Mean	**1.968**	**0.309**
	Median	1.190	0.200
ASH	Mean	**1.856**	**0.279**
	Median	1.124	0.190
ASHD	Mean	**0.112**	**0.022**
	Median	0.062	0.011
TOC	Mean	**0.258**	**0.076**
	Median	0.010	0.000
CNE	Mean	**0.168**	**0.095**
	Median	0.100	0.000
CPN	Mean	**0.238**	**0.056**
	Median	0.100	0.000
LEN	Mean	**0.685**	**0.125**
	Median	0.430	0.070

Table 2 provides example scores for a comment. **Table 3** presents mean and median feature values across all features for both classes.

3. Preliminary Experiments

3.1. Technical Details of Initial Network Models

As classifiers, we used both shallow and deep networks for comparison.

For a shallow network, we used our model from [18]. That model had an input layer of one unit per each of our 14 features, a hidden layer of 10 units using the tanh activation function and an output layer with one unit using the sigmoid activation function. For training, we used the parameters learning rate = 0.2 and number of epochs = 20,000. The weights were initialized randomly.

We also decided to use a deep neural network with longer training time to potentially reduce avoidable bias. To find the best hyperparameters, we conducted hyperparameter search using the fine-to-coarse search method. Essentially, at first we randomized the hyperparameter values between two set numbers and in the following rounds we kept narrowing this interval with respect to values that showed best performance. We tuned five hyperparameters in total: the initial learning rate α, the number of epochs, the number of layers, the number of neurons in each layer and the mini-batch size for gradient descent. The β_1, β_2, and epsilon values for Adam [19] were set at 0.9, 0.999, and 10^{-8} respectively. The activation function was tanh for all layers except for the output layer, which was a single sigmoid unit.

We chose the model that performed best on our dataset. The initial. α value for this model was 0.0037. The network had 3 hidden layers with 7, 8, and 15 units, respectively. It was trained for 30,000 epochs using the batch gradient descent method. Weights for all the deep network models described above were initialized using the Xavier method [20].

3.2. Experiments

Using our *r/getdisciplined* vs. *r/todayilearned* datasets, we performed initial experiments until we arrived at our proposed method of classification. All experiments were performed as 10-fold cross validation. For each fold, the training set had 1,600 examples and the test set had 179 examples.

Although our 14 advice features performed well enough in our previous experiments on this topic, this time we used Word2Vec [21] vectors to increase the number of features. We obtained the vectors for each text using a model trained on texts from Wikipedia (latest version downloaded in February 2018 from [22]). We used the skip-gram method, vectors of size = 100, minimum word frequency = 5 and number of epochs = 10. All hyperparameters not specified above were left at default values. We trained two models: one on words and one on single characters. We used the character model to deal with unknown words. Specifically, a vector for the unknown word was calculated as an average of all character vectors of this word. For each text, we calculated the text vector by averaging all sentence vectors of that text, and we calculated the sentence vectors using the method described in [23]. Text vector was then calculated as average of all sentence vectors in the comment.

Before calculating the features, we performed spellchecking and contraction expanding on each comment text using code implementations available online from [24] and [25] respectively. Moreover, we deleted all non-ASCII characters from the comment texts.

There were two feature sets: one consisting only of Word2Vec text vectors and one where those vectors were extended with our 14 features. The vector features were normalized using L2 norm. The 14 advice features were left unchanged.

3.3. Results

Results are presented in **Table 4**. *Shallow* and *Deep* mean our shallow and deep models, respectively. *100 features* denotes experiments where only 100-dimensional Word2Vec vectors were used as features. *114 features* means experiments where we concatenated those word vectors with our own 14 advice features. The *14 features* is a shallow model trained using only our 14 advice features and is included for comparison. *Acc*, *prec*, and *rec* denote accuracy, precision, and recall, respectively. We show results up to the third decimal point to better display the minute differences between the models. We bolded best results in each column.

The experiment results showed that shallow networks worked better for this task and that whether we incorporated our 14 features or not did not make any difference when using Word2Vec features. This can be explained mathematically by the way neural networks work. In fully connected networks, each layer's activation values are computed by multiplying the previous layer's activations by the current layer's weights, adding a bias term and putting the results through a non-linear function. Each neuron from the current layer is connected to each neuron in the previous layer. This means that each neuron in the current layer takes as input all values from the layer before it. In our case, this means that the first hidden layer of our network took all 100 or all 114 feature values. It is possible that compressing all these values into each single neuron in the hidden layer erased the significance of the 14 additional features. In other words, as a feature number, 100 was not significantly different from 114. Moreover, perhaps because of the lack of normalization on the 14 advice features, the results where we used both word embeddings and our 14 features were slightly worse that with only Word2Vec vectors.

However, previous experiments already proved that our 14 features were very informative for the neural network performing our classification task. Word embeddings by themselves also gave us good results. Therefore, we decided to employ a novel, original network architecture to leverage the strengths of both feature sets while avoiding overshadowing one by the other. We propose an approach to classification where we use two shallow networks, one trained solely on vector embeddings and one trained solely on 14 advice features. However, to use both

Table 4. Results from experiments using Word2Vec word embeddings.

	Training set				Test set			
	Acc	Prec	Rec	F-score	Acc	Prec	Rec	F-score
Shallow 100 features	0.948	0.943	0.953	0.948	**0.912**	**0.910**	**0.916**	**0.912**
Shallow 114 features	0.948	0.949	0.946	0.948	0.904	0.906	0.900	0.903
Shallow 14 features	0.886	0.898	0.868	0.883	0.865	0.880	0.847	0.863
Deep 100 features	**0.999**	**1.000**	**0.998**	**0.999**	0.880	0.877	0.884	0.880
Deep 114 features	**0.999**	**1.000**	**0.998**	**0.999**	0.898	0.893	0.905	0.899

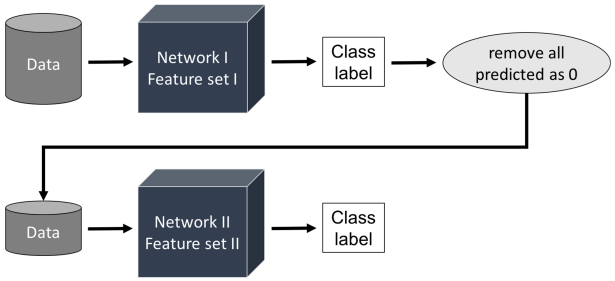


Fig. 2. Illustration of the proposed method.

of these feature sets in classification, we connect the two networks by using the output of the word embeddings network for creating input to the advice features network. From this point on, we will call them *Embeddings Network* and *Advice Network* respectively.

4. Proposed Method

4.1. Network Architecture

We propose to perform the classification twice, each time using a different network with a different feature set. By different feature sets, we mean the 100 Word2Vec features as one set and the 14 advice features as the other set. The Embeddings Network uses Word2Vec features while the Advice Network uses the 14 advice features. **Fig. 2** illustrates the general flow of data through both networks.

As seen in **Fig. 2**, there are two stages of classification in this method. The first network classifies data according to the results of its training. After this first stage is performed, each data example has a predicted label 1 (meaning *motivational*) or 0 (meaning *non-motivational*). Then all examples classified by the first network as 0 are **removed** from the dataset. While this may mean throwing away valuable data that was misclassified, it should also significantly reduce the amount of data irrelevant to the goal of this research. In other words, it should remove regular texts that do not contain any advice. After the removal, this filtered dataset is then input into the second network, which confirms the predicted label as 1 or classifies the text as 0, contrary to the first network. Only the texts classified twice as 1 are regarded as *motivational* at the end.

Table 5. Data for testing the proposed method.

Class	Subreddit	Number of comments
Motivational	*r/getdisciplined*	279
Non-motivational	*r/todayilearned*	300
	Total	579

Figure 2 contains generic names like *Network I* and *Network II*. This is because either network – *Embeddings* or *Advice* – could be the first classifier and the other one could be the second, since both networks performed similarly well on the 10-fold cross validation experiments (as seen in **Table 4**). We performed experiments with both network orders to see whether there was any difference between them.

4.2. Experiments

For both networks, we used the models that were already trained and that were described in the previous section. For *Embeddings Network*, it was the 10 models trained as *Shallow 100 features*. For *Advice Network*, it was the 10 models trained as *Shallow 14 features*. All the hyperparameters and data was the same as described in the previous Section.

The data that was used for testing was an entirely new dataset downloaded from *r/getdisciplined* and *r/todayilearned*. **Table 5** presents the data.

For each data example, we obtained a 114-feature vector, where the first 100 features were word embeddings and the last 14 features were the advice features. Each data entry was input to the two classifiers with its corresponding feature set. For example, if the first classifier was the *Embeddings Network*, then the input was the 100 Word2Vec features. Then, if the text was classified as 1, it was passed to the *Advice Network*, this time using its 14 advice features.

We performed two experiments. In the first one, we chose the best *Embeddings Network* model as the first classifier. This model was chosen based on its performance on the test set in the 10-fold cross validation training with regards to F-score. Therefore, the test sets for the 10 models essentially became our development sets. During the second stage of classification, we input the results

from *Embeddings Network* to each of the 10 models of *Advice Network* and obtained mean accuracy, precision, recall, and F-score.

The second experiment switched the networks to see whether the order of the networks influences the results. This time the first classifier was the best model of *Advice Network* and the second classifiers were the 10 models of *Embeddings Network*.

4.3. Results

Results of the two experiments are presented in **Tables 6** and **7**, respectively. We tested each model on each step of the way. The first row in each Table presents results for the chosen first classifier – the best *Embeddings Network* or *Advice Network* model – on its previous test set. The second row presents its results on the current test set shown in **Table 5**. Row number three shows results on that test set after removing comments predicted as 0. Obviously, recall for this row is 1.0, since we removed one predicted class, effectively getting rid of both true negatives and false negatives. Accordingly, the F-score rises as well. Therefore, this row is presented only for reference purposes. Row number four shows mean results for the 10 models that served as the second classifier.

With the experiment where *Embeddings Network* was the first classifier and *Advice Network* was the second one, there were 289 out of total 579 examples left in the dataset after removing predicted class 0 (meaning *non-motivational*). There were 22 examples in that removed data that were misclassified as 0 but actually had the true label 1 (meaning *motivational*). This means 7.88% loss of *motivational* data during the classification. In the experiments where *Advice Network* was the first classifier followed by 10 models of *Embeddings Network*, there were 267 examples left in the data after the removal. 34 advisory/motivational examples were lost in that removal, which is 12.18% of the class 1 dataset.

4.4. Comparison with a Pool of Voters

To confirm the effectiveness of our method, we compared it with a pool of voters. This was done to see how the proposed method fares against other methods utilizing multiple networks.

Each pool was comprised of individual models trained in the 10-fold cross-validation fashion, which gave us 10 models per pool. Just as with the proposed method, all models were trained on the *r/getdisciplined* and *r/todayilearned* data shown in **Table 1** and tested on the unseen data shown in **Table 5**.

There were 4 conditions:

1. Shallow network models trained with only the 100 Word2Vec features (10 models).
2. Shallow network models trained with only the 14 advice features (10 models).
3. Models from Condition 1 and Condition 2 pooled together (20 models). This was done to see whether the models trained separately on the two feature sets would be able to act as one pool and overcome the proposed method by voting together.
4. Shallow network models trained with all 114 features (10 models).

In other words, models from Conditions 1 and 2 were the same models used in our proposed method. Conditions 3 and 4, on the other hand, combined the two feature sets in different ways to see how that compares to our combination of *Embeddings* and *Advice Networks*.

As will be discussed in Section 5.2, usually not all voters in the pool have the same weights. Based on performance, some of them get more significant votes than others. To reflect that, we did not use all voters in the pool. Instead, we used only top 9, 5, 3, and 1 for each condition in four separate experiments. We chose odd numbers so that there would always be a majority in voting. For each condition, we ranked the models based on their performance on the test set during the 10-fold cross validation training with respect to F-score. Then, these models were tested on the new dataset that was presented in **Table 5**. As the final prediction, we chose the class that most models voted on. We then compared this prediction against the true label and calculated all metrics from that.

For comparison, we also calculated the mean results of all models for each condition except Condition 3, which were just models from Condition 1 and Condition 2 pooled together. Results are presented in **Table 8**.

4.5. Discussion About Proposed Method

As evident from **Tables 6** and **7**, we were able to improve on the results provided by either feature set alone. The F-score rose to 0.94 with the *Embeddings-Advice* order and to 0.97 with the *Advice-Embeddings* order. Moreover, the method beats all combinations of pool of voters as well as the 114 features used together, as can be seen in **Table 8**. The results are especially important when compared with the pool of voters. Since the *14 features* models performed worse overall than the *100 features* models (as seen in **Table 4**), they were outranked in the pool of voters in Condition 3 and we could not utilize their strengths. On the other hand, our method puts proper attention on both feature sets and as such, proved to be more effective for classifying texts into motivational/advisory and regular categories.

It is debatable which order of networks is better. Results were significantly higher for the *Advice-Embeddings* order. However, that experiment meant the loss of almost 13% of relevant data. In the other experiment the loss was only around 8%, but F-score was lower by 3% and accuracy by as much as 9%.

Since our main goal was to extract advisory texts from an unknown dataset, the metrics that matters most is precision. Precision is defined as the ratio of true positives to all positives. In other words, it shows how many comments predicted as motivational/advisory truly belonged to this class. The higher the precision, the less noise there

Table 6. Results for the order *Embeddings Network – Advice Network*.

Step	Accuracy	Precision	Recall	F-score
Previous test set	0.94	0.92	0.95	0.93
Current test set	0.89	0.89	0.92	0.90
After removal	0.88	0.89	1.00	0.94
Mean for Advice Networks	0.84	0.97	0.92	0.94

Table 7. Results for the order *Advice Network – Embeddings Networks*.

Step	Accuracy	Precision	Recall	F-score
Previous test set	0.90	0.89	0.90	0.90
Current test set	0.83	0.92	0.88	0.90
After removal	0.84	0.92	1.00	0.96
Mean for Embeddings Networks	0.93	0.98	0.96	0.97

Table 8. Results for the experiments involving a pool of voters. For the *Both 100 and 14 features condition*, which included 20 models (10 from *100 features* and 10 from *14 features*) all best models came from the *100 features* condition, which makes their results identical. This is indicated by the *Same as 100* label. Best results in each row are bolded.

	Condition	100 features	14 features	Both 100 and 14 features	114 features
Best 9	Accuracy	**0.921**	0.903	Same as 100	0.919
	Precision	**0.921**	0.914	Same as 100	0.920
	Recall	**0.914**	0.882	Same as 100	0.910
	F-score	**0.918**	0.898	Same as 100	0.915
Best 5	Accuracy	**0.919**	0.905	Same as 100	0.914
	Precision	0.911	**0.924**	Same as 100	0.916
	Recall	**0.921**	0.875	Same as 100	0.903
	F-score	**0.916**	0.899	Same as 100	0.910
Best 3	Accuracy	**0.917**	0.900	Same as 100	0.914
	Precision	**0.914**	**0.914**	Same as 100	0.910
	Recall	**0.914**	0.875	Same as 100	0.910
	F-score	**0.914**	0.894	Same as 100	0.910
Best 1	Accuracy	0.907	0.903	Same as 100	**0.919**
	Precision	0.889	0.918	Same as 100	**0.920**
	Recall	**0.921**	0.878	Same as 100	0.910
	F-score	0.905	0.897	Same as 100	**0.915**
All (mean)	Accuracy	**0.896**	0.828	-	0.895
	Precision	**0.914**	0.911	-	**0.914**
	Recall	**0.911**	0.870	-	0.909
	F-score	**0.912**	0.890	-	0.911

is in the extracted data. Precision was high in both experiments, reaching 97% or 98%. This is the main reason that our method is effective: it reduces the noise to 2–3%.

However, there was some loss in data when removing the comments classified as regular by the first classifier. It seems to be a trade-off with precision, since the order of networks that gave lower precision also removed fewer relevant comments than the other one. It remains to be seen whether this trade-off would appear consistently across multiple experiments with various datasets and networks with different parameters. The results presented here mean that the effectiveness of this method depends on what is most important to the goal of the research. Moreover, the order of the networks, if computationally feasible, should be determined experimentally and not in an arbitrary manner.

The proposed method needs further exploration in other areas as well.

First, it may be effective for other text classification problems as well. Semantic word embeddings have proven to be useful in all kinds of textual data processing. Normally, they are concatenated with any additional features used in experiments, but simple concatenation may obscure the significance of those additional features by processing them together with all the numerous vector values. Our method alleviates this problem by giving proper attention to each feature set by using them as input to their own separate neural networks. Perhaps the same method can be used for other text classification tasks where input features consist of both word vectors and original features devised for the task at hand. However, since the method is not dependent on dataset qualities or feature qualities, it may be possible to use it for tasks in different domains as well.

Secondly, it is important to note here that both networks were solving the same classification task and that the two feature sets were not entirely independent of each other. Both the 100 Word2Vec features and the 14 advice features were calculated based on the same text, so while they emphasized different qualities of that text, some information may have been shared between the sets, even if not in a readily visible way. It is possible that connecting the two networks allowed us to approach the same text qualities in two different, mutually complementing ways. At this moment, it is not possible to say whether this relationship between feature sets is relevant to the results or not.

Thirdly, the method described in this paper was devised for binary classification. It might be interesting to create a similar method for multiple classes. Further studies on this topic could include exploring the relationship between final results and data loss for each relevant class.

Last but not least, we were able to leverage both methods in this way because both networks worked well enough by themselves. If any of the networks had given lesser accuracy, for example around 0.7, this probably would have badly influenced the final results. This means that our method may only be useful in settings where both feature sets do well on their own, or at least one of them makes up for the shortcomings of the other.

There are various benefits to the proposed method, like its simplicity, predictable computational cost (which is the combined cost of all networks involved) and apparent effectiveness. However, there is still room for creative improvement. For example, even though in this case we used the same network architecture for both networks, this is not obligatory. Since both networks are trained separately, their hyperparameter values can be easily readjusted on a network basis to best fit the purposes of the research. Moreover, perhaps more than two networks can be chained to gradually increase the accuracy of the last one.

It is possible that we would have obtained even better results, had we used Word2Vec vectors with more dimensions than 100 (for example 300 or 500). It is generally expected that a neural network should work better when it has more features to compute. However, it is also important to think of the computational cost. Increasing the number of Word2Vec features to 300 or 500 would mean increasing the number of network parameters as well. There is a trade-off in deep learning between using as few network parameters as possible while still obtaining good results. We believe we had that balance in our experiments. The combined networks already performed with F-score of 94% or more, so increasing the number of Word2Vec features was not likely to bring much improvement. On the contrary; more computations also mean that the network could overfit to the training set, bringing test set results down. This was shown with our deep models that had more layers and more hidden units (see **Table 4** for comparison of training set and test set results).

5. Related Work

5.1. User Motivation

While there exist papers suggesting various approaches to influencing motivational states in users, they do not contain experiments confirming their hypotheses. Authors of [26] propose a model for characterizing user motivation by applying psychological research, for example Attribution Theory [27], which stipulates that the outcomes of an individual's actions influence both what that individual perceives as contributing factors to the actions, and the emotional responses to those actions. Authors of [28] describe a virtual agent that uses various motivational strategies aimed at PTSD patients, based on their therapy progress and trust in success of the treatment. [29] describes research on types of feedback that would be effective for people at different stages of readiness, willingness and ability to change in motivating them to save electrical energy. In [30] a model is constructed for user motivation in human-robot cooperation to help the user achieve daily physical activity goals. The authors propose an adaptive robot companion that models the user's motivation daily and changes its strategies accordingly. [31] presents a user behavior model for developing persuasive technologies, defined with three components: motivation, ability and triggers. In the author's view, target behavior takes place where both motivation and ability to perform the task are high and triggers are present. [32] describes the basics for creating persuasive systems aiming to change the user's behavior. As a core construct, the author proposes an outcome/change matrix for desired actions, which should serve as guide for determining design goals for behavior change support systems. All these papers present frameworks, but do not verify their claims through experiments. Moreover, they do not deal with any aspect of natural language processing. Those that do use motivating feedback in the form of text or utterances only generally describe what should be included in that feedback, with no generative process to speak of. Finally, the proposed frameworks are either too loosely related to our research problem ([26, 31, 32]) or too specific in their goals ([28–30]) to be adapted as basis for our studies.

Empirically verified studies include motivating users to do indoor-cycling every day for a specified period of time with a robot companion by building and maintaining a human-computer relationship [33] or encouraging users to perform longer planking exercise by giving them acknowledging feedback from a robot that exercised together with them [34]. However, in both cases the dialogue with the agent was scripted and limited to a few topics. Moreover, both studies dealt only with exercise and their very specific results cannot be generalized to other everyday tasks. Authors of [35] describe a study where subjects were being persuaded to reduce snacking via personalized short text messages. The messages were tailored to the user based on the user's score on the Susceptibility to Persuasion Scale [36]. However, these messages were all crafted by the researchers and involved no natural language processing.

5.2. Multiple Networks for Data Classification

The approach proposed in this paper is not the first to use multiple neural networks in classification. For example, the authors of [37] propose a general method of classification where they combine output from multiple networks trained on the same data to decide on the final result. This is done via fuzzy membership function, in which each network has its own weight in the final decision. Jerebko, Malley, Franaszek, and Summers [38] describe a method of classifying CT scans for presence of colonic polyps in radiology, where they determine 12 features and distribute their random subsets between multiple networks. Each network computes its own results and gets to vote for the final decision, with a weight being given to it based on its performance levels. Tsuda, Shin, and Schölkopf [39] show a method of combining multiple networks for protein classification and determining those that were the most important for obtaining good results. What all these papers have in common is that they see multiple networks as a pool of voters on the final decision. In a way, our method can also be regarded as voting, except that the networks do not vote together. Instead, they are connected in a linear way where the second network only works with the examples that were voted on as class 1 by the first network. However, our experiments showed that our method yields better performance than a simple pool of voters.

More than one network is also used in the technique called Generative Adversarial Networks [40], however in that case both networks have different purposes and compete with each other. This is different from our goal, because we want both networks to work together on solving the exact same task.

Siamese networks [41] for face verification also employ two neural networks. They share the weights and architecture, and give outputs of identical type on two different input images, which is then used for final verification. However, we do not compare outputs of our networks, at least not in a straightforward way; doing that would be similar to actual voting. Secondly, as mentioned in Section 4.5, there is no obligation for both networks to be identical in our method.

Gkioxari, Toshev, and Jaitly [42] presented a method of chaining convolutional neural networks for the task of human pose prediction from images. They implemented an architecture similar to sequence-to-sequence models, where the output of each model depended on its input combined with the outputs of previous models. In this method, the outputs are plugged as a part of input to each subsequent network. In our experiments, we used the output of one network to clean the data that is the input to the second network. This is different than actually inputting the class label provided by the first network to the second one.

6. Conclusions

In this paper, we have described our research on distinguishing motivational advice from regular texts. We have successfully determined 14 features for recognition of motivational/advisory texts from regular ones. In our first experiment, described in [18], we achieved accuracy of 0.88 on the test set using a shallow neural network with F-score of 0.76. After performing numerous other experiments in attempts to raise the final scores, we discovered a novel method for text classification by chaining two shallow networks together. In the end, both F-score improved to 0.94 or 0.97.

The two feature sets that we have used for classification were our original 14 advice features and word embeddings of length 100. Each feature set performed reasonably well on its own as input to the classifier. This shows that motivational/advisory texts are different from non-advisory ones not only in visible ways like the former using advisory and imperative expressions while the latter does not, but also on a basic semantic level expressed in Word2Vec vector elements. Using both feature sets allowed us to capitalize on these primitive semantic qualities not readily visible to the human eye as well as on the strengths of features designed manually after looking into the data. Moreover, chaining both networks made it possible to combine both feature sets in a unique way, which translated not only to better results, but also to better understanding of the qualities that motivational/advisory texts possess. This means that in the future stages of our research, when generating textual motivational advice for the user, we may have to take both these feature sets into consideration. While vector embeddings are typically used in recurrent neural networks, there is a need to devise an effective way of incorporating our 14 advice features as well, perhaps by letting them guide the Beam Search algorithm.

The methods described in this paper will allow us to carefully choose the training data for our end-goal dialogue system. Therefore, these experiments are a crucial stage in constructing a motivational agent, which is the final goal of our research.

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